Desenhe os gráficos de Bode para

$$\mathbf{H}(s) = \frac{s + 1}{s^2 + 12s + 100}$$

$$H(\omega) = H(s)\Big|_{s=j\omega} = \frac{j\omega + 1}{(j\omega)^2 + 12j\omega + 100}$$

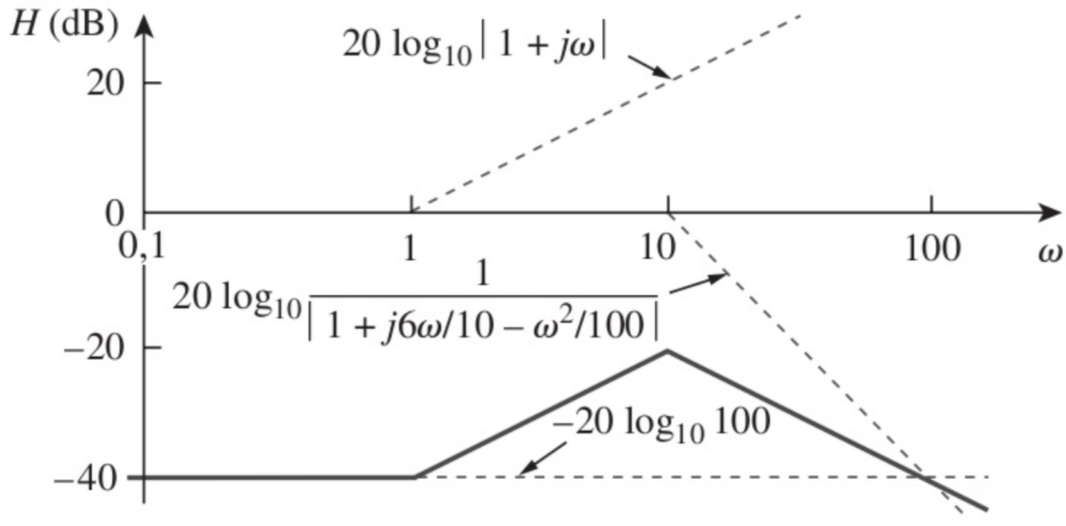
$$= \frac{j\omega + 1}{100\left[1 + \frac{12j\omega}{100} + \frac{(j\omega)^2}{100}\right]}$$

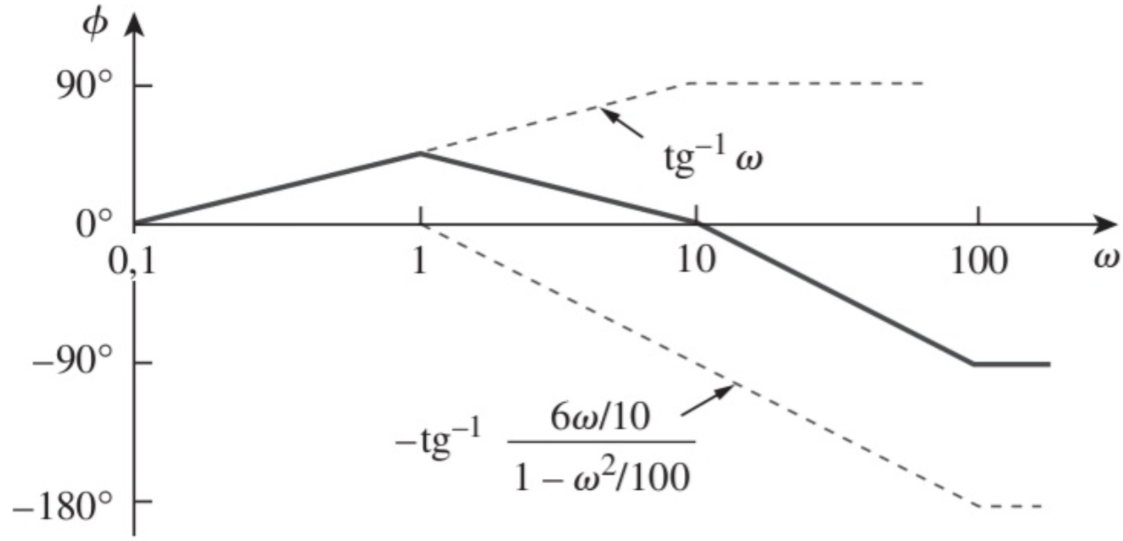
$$= \frac{1/100\,(j\omega + 1)}{1 + j\omega\frac{1{,}2}{10} + \left(\frac{j\omega}{10}\right)^2}$$

$$\mathbf{H}(\omega) = \frac{1/100(1 + j\omega)}{1 + j\omega 1{,}2/10 + (j\omega/10)^2}$$

$$H_{\text{dB}} = -20\log_{10} 100 + 20\log_{10}|1 + j\omega| - 20\log_{10}\left|1 + \frac{j\omega 1{,}2}{10} - \frac{\omega^2}{100}\right|$$

$$\phi = 0° + \text{tg}^{-1}\,\omega - \text{tg}^{-1}\left[\frac{\omega 1{,}2/10}{1 - \omega^2/100}\right]$$





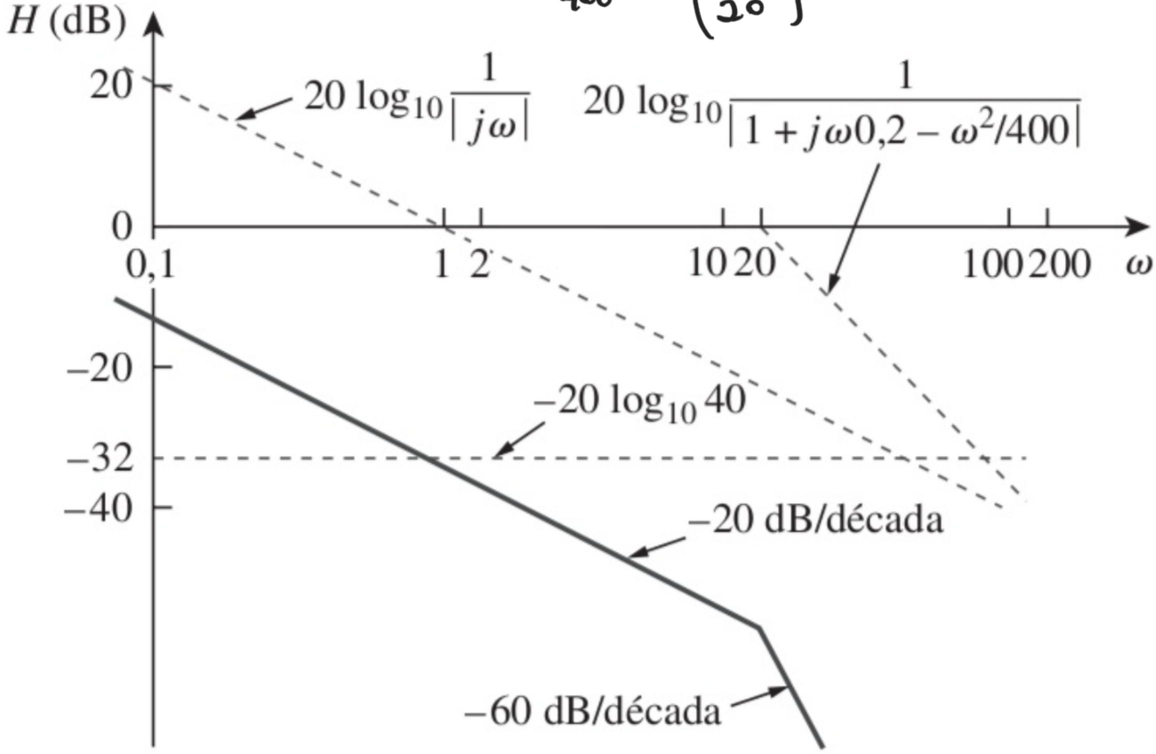
$\phi$
90°
0°
−90°
−180°
0,1
1
10
100
$\omega$
$\mathrm{tg}^{-1}\,\omega$
$-\mathrm{tg}^{-1}\ \dfrac{6\omega/10}{1-\omega^2/100}$

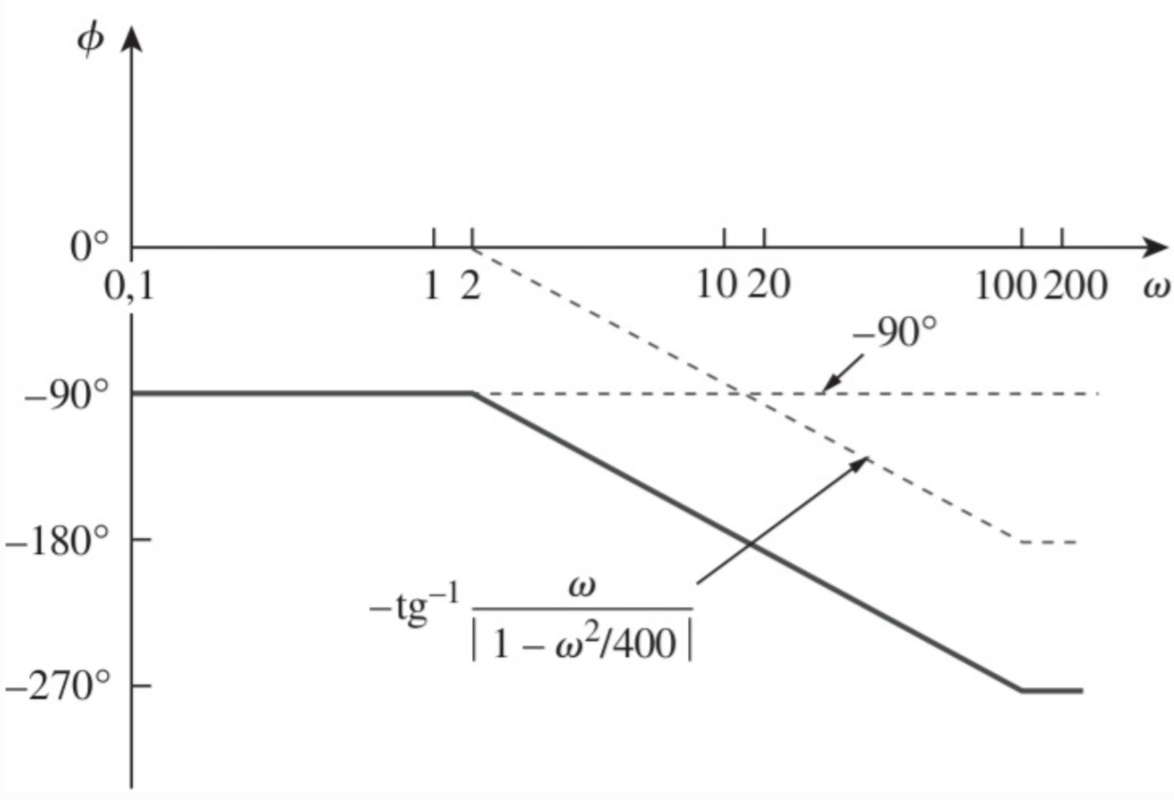
Construa os gráficos de Bode para

$$H(s) = \frac{10}{s(s^2 + 80s + 400)}$$

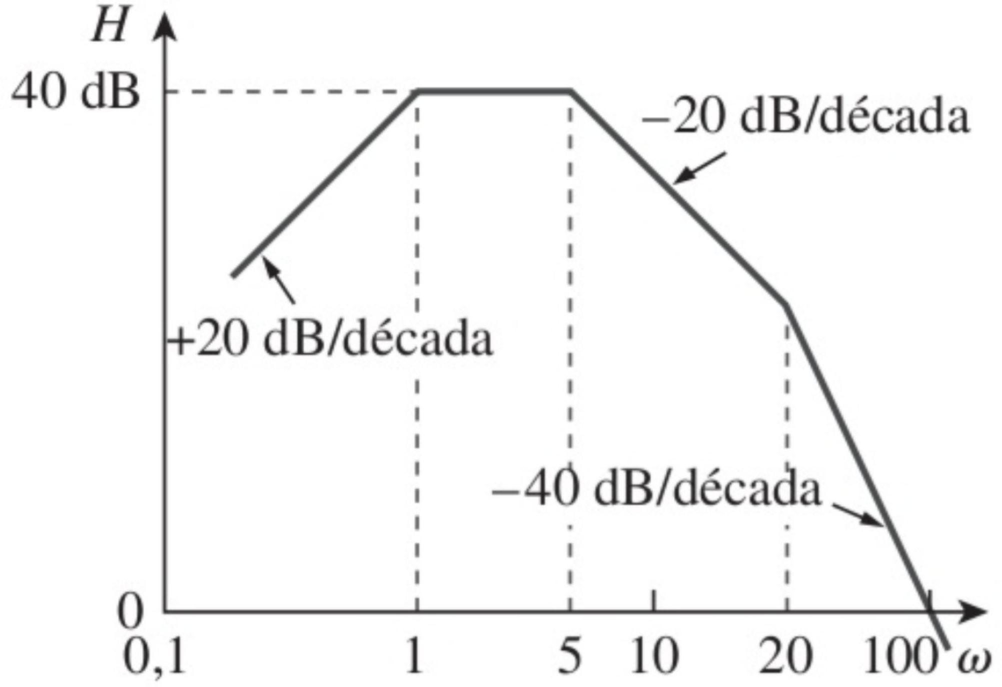
$$H(\omega) = H(s)\Big|_{s=j\omega} = \frac{10}{j\omega\left[(j\omega)^2 + 80j\omega + 400\right]}$$

$$= \frac{1/40}{j\omega\left[1 + \frac{80}{400}j\omega + \left(\frac{j\omega}{20}\right)^2\right.}$$



Dado o gráfico de Bode da Figura, obtenha a função de transferência $\mathbf{H}(\omega)$.



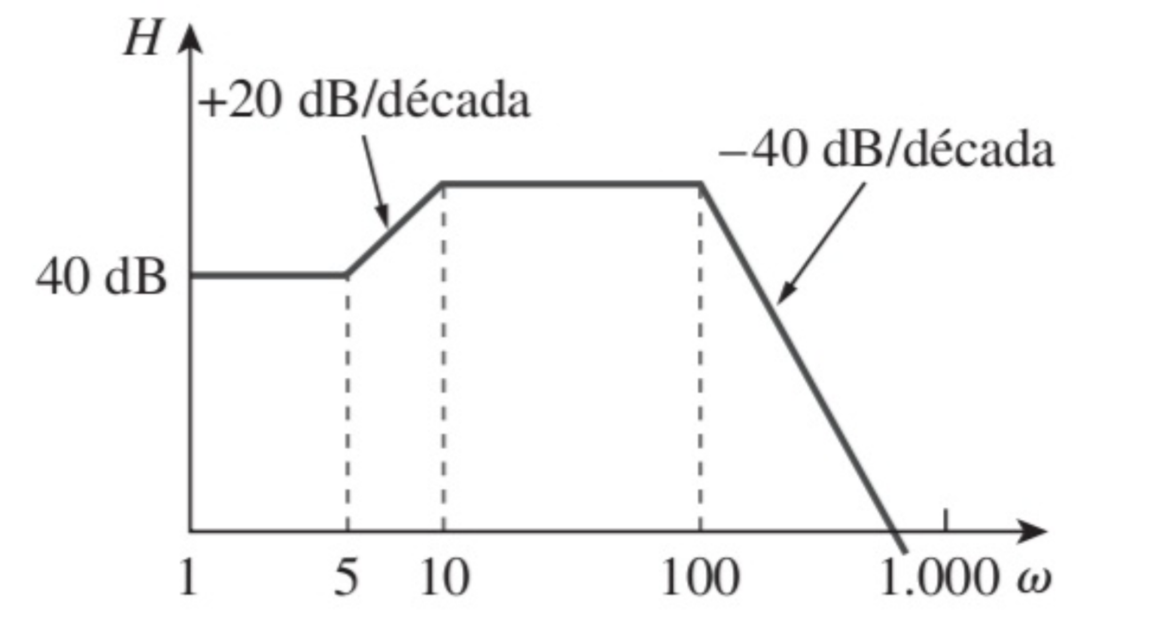

$$40 = 20 \log_{10} K \qquad \Rightarrow \qquad \log_{10} K = 2$$

$$K = 10^2 = 100$$

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{100 j\omega}{(1 + j\omega/1)(1 + j\omega/5)(1 + j\omega/20)} \\ &= \frac{j\omega 10^4}{(j\omega + 1)(j\omega + 5)(j\omega + 20)}\end{aligned}$$

$$\mathbf{H}(s) = \frac{10^4 s}{(s + 1)(s + 5)(s + 20)}, \qquad s = j\omega$$

Obtenha a função de transferência $\mathbf{H}(\omega)$ correspondente ao gráfico de Bode na Figura




**Resposta:** $\mathbf{H}(\omega) = \dfrac{2.000.000\,(s + 5)}{(s + 10)(s + 100)^2}$.